Name: Digvijay Vikas Sawant

PRN: 260240125023

Form no: 251100327

Q1. Use `pwd`, `ls`, `cd` to explore the file system and display current location.

```
digvijay@digvijay-virtualbox: ~/bda
digvijay@digvijay-virtualbox:~$ pwd
/home/digvijay
digvijay@digvijay-virtualbox:~$ ls
bda       Documents  Music     Public  Templates
Desktop   Downloads  Pictures  snap    Videos
digvijay@digvijay-virtualbox:~$ cd bda/
digvijay@digvijay-virtualbox:~/bda$
```

Q2. Create directories `lab1/lab2/lab3` using one command.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_1
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1$ mkdir -p lab1/lab2/lab3
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1$ ls
lab1
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1$ tree
.
└── lab1
    └── lab2
        └── lab3

4 directories, 0 files
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1$
```

Q3. Create 10 files (`file1` to `file10`) inside `lab3` using one command.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_1/lab1/lab2/lab3
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1$ cd lab1/lab2/lab3/
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2/lab3$ touch file{1..10}
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2/lab3$ ls
file1  file10  file2  file3  file4  file5  file6  file7  file8  file9
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2/lab3$
```

Q4. Display who is logged in with `who`, and the current user with `whoami`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_1/lab1/lab2/lab3
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2/lab3$ who
digvijay tty2         2026-02-27 20:57 (:0)
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2/lab3$ whoami
digvijay
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2/lab3$
```

Q6 Create 50 files `data1` to `data50` inside a directory with a single command.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_1/lab1
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1$ touch data{1..50}
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1$ ls
data1   data13  data17  data20  data24  data28  data31  data35  data39  data42  data46  data5   data8
data10  data14  data18  data21  data25  data29  data32  data36  data4   data43  data47  data50  data9
data11  data15  data19  data22  data26  data3   data33  data37  data40  data44  data48  data6   lab2
data12  data16  data2   data23  data27  data30  data34  data38  data41  data45  data49  data7
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1$
```

Q7. Copy all `.txt` files from one folder to another using wildcards.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_1/lab1/lab2
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2/lab3$ ls
file1  file10  file2  file3  file4  file5  file6  file7  file8  file9  text1.txt  text2.txt  text3.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2/lab3$ cp ~/Assignment/Assignment_1/lab1/lab2/lab3/*.tx
t ~/Assignment/Assignment_1/lab1/lab2
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2/lab3$ cd ..
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2$ ls
lab3  text1.txt  text2.txt  text3.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2$
```

Q8. Use `cat`, `less`, `head`, `tail` to view file contents.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_1/lab1/lab2
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2$ nano text1.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2$ cat text1.txt
abcd
efgh
ijkl
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2$ head -n 1 text1.txt
abcd
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2$ tail -n 1 text1.txt
ijkl
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2$ less text1.txt
```

```
abcd
efgh
ijkl
text1.txt (END)
```

Q10. Redirect the output of `ls -l` to a file and append the date to it.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_1/lab1/lab2
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2$ { ls -l; date; } >> text2.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2$ cat text2.txt
-rw-rw-r-- 1 digvijay digvijay 15 Feb 27 21:56 text1.txt
total 12
drwxrwxr-x 2 digvijay digvijay 4096 Feb 27 21:48 lab3
-rw-rw-r-- 1 digvijay digvijay  295 Feb 27 23:22 text1.txt
-rw-rw-r-- 1 digvijay digvijay   57 Feb 27 22:09 text2.txt
-rw-rw-r-- 1 digvijay digvijay    0 Feb 27 21:50 text3.txt
Friday 27 February 2026 11:23:23 PM IST
digvijay@digvijay-virtualbox:~/Assignment/Assignment_1/lab1/lab2$
```

Name: Digvijay Vikas Sawant

PRN: 260240125023

1. Create a directory named `workspace` in your home folder.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~$ mkdir workspace
digvijay@digvijay-virtualbox:~$ ls
Assignment  bda  Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos  workspace
digvijay@digvijay-virtualbox:~$ cd workspace/
digvijay@digvijay-virtualbox:~/workspace$
```

2. Inside `workspace`, create two subdirectories named `source` and `backup`.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ mkdir source backup
digvijay@digvijay-virtualbox:~/workspace$ ls
backup  source
digvijay@digvijay-virtualbox:~/workspace$
```

3. Create three empty files in `source` named `alpha.txt`, `beta.txt`, and `gamma.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace/source
digvijay@digvijay-virtualbox:~/workspace/source$ touch alpha.txt beta.txt gamma.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls
alpha.txt  beta.txt  gamma.txt
digvijay@digvijay-virtualbox:~/workspace/source$
```

4. Rename `alpha.txt` to `delta.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace/source
digvijay@digvijay-virtualbox:~/workspace/source$ ls
alpha.txt  beta.txt  gamma.txt
digvijay@digvijay-virtualbox:~/workspace/source$ mv alpha.txt delta.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls
beta.txt  delta.txt  gamma.txt
digvijay@digvijay-virtualbox:~/workspace/source$
```

5. Move `delta.txt` from the `source` directory to the `backup` directory.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/source$ ls
beta.txt  delta.txt  gamma.txt
digvijay@digvijay-virtualbox:~/workspace/source$ mv delta.txt ~/workspace/backup/
digvijay@digvijay-virtualbox:~/workspace/source$ cd ..
digvijay@digvijay-virtualbox:~/workspace$ cd backup/
digvijay@digvijay-virtualbox:~/workspace/backup$ ls
delta.txt
digvijay@digvijay-virtualbox:~/workspace/backup$
```

6. Create a new user named `intern`.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/backup$ sudo adduser intern
[sudo] password for digvijay:
info: Adding user `intern' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `intern' (1001) ...
info: Adding new user `intern' (1001) with group `intern (1001)' ...
info: Creating home directory `/home/intern' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for intern
Enter the new value, or press ENTER for the default
        Full Name []: intern
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `intern' to supplemental / extra groups `users' ...
info: Adding user `intern' to group `users' ...
digvijay@digvijay-virtualbox:~/workspace/backup$
```

7. Create a new group named `training`.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/backup$ sudo groupadd training
digvijay@digvijay-virtualbox:~/workspace/backup$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,digvijay
```

```
vboxsf:x:123:
intern:x:1001:
training:x:1002:
digvijay@digvijay-virtualbox:~/workspace/backup$
```

8. Add the user `intern` to the `training` group.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/backup$ sudo usermod -aG training intern
digvijay@digvijay-virtualbox:~/workspace/backup$ groups intern
intern : intern users training
digvijay@digvijay-virtualbox:~/workspace/backup$
```

9. Change the password for the user `intern`.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/backup$ sudo passwd intern
New password:
Retype new password:
passwd: password updated successfully
digvijay@digvijay-virtualbox:~/workspace/backup$
```

10. Create a file named `private.txt` and set permissions so only the owner can read and write to it ($rw-------$).

```
digvijay@digvijay-virtualbox: ~/workspace/source
digvijay@digvijay-virtualbox:~/workspace/source$ touch private.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls -l private.txt
-rw-rw-r-- 1 digvijay digvijay 0 Feb 27 22:52 private.txt
digvijay@digvijay-virtualbox:~/workspace/source$ sudo chmod 600 private.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls -l private.txt
-rw------- 1 digvijay digvijay 0 Feb 27 22:52 private.txt
digvijay@digvijay-virtualbox:~/workspace/source$
```

11. Create a file named `public.txt` and set permissions so everyone can read it, but only the owner can write to it ($rw-r--r--$).

```
digvijay@digvijay-virtualbox: ~/workspace/source
digvijay@digvijay-virtualbox:~/workspace/source$ touch public.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls -l public.txt
-rw-rw-r-- 1 digvijay digvijay 0 Feb 27 22:56 public.txt
digvijay@digvijay-virtualbox:~/workspace/source$ sudo chmod 644 public.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls -l public.txt
-rw-r--r-- 1 digvijay digvijay 0 Feb 27 22:56 public.txt
digvijay@digvijay-virtualbox:~/workspace/source$
```

12. Create a file named `script.sh` and give it execute permissions for the owner, group, and others ($rwxr-xr-x$).

```
digvijay@digvijay-virtualbox: ~/workspace/source
digvijay@digvijay-virtualbox:~/workspace/source$ touch script.sh
digvijay@digvijay-virtualbox:~/workspace/source$ ls -l script.sh
-rw-rw-r-- 1 digvijay digvijay 0 Feb 27 22:57 script.sh
digvijay@digvijay-virtualbox:~/workspace/source$ sudo chmod 755 script.sh
digvijay@digvijay-virtualbox:~/workspace/source$ ls -l script.sh
-rwxr-xr-x 1 digvijay digvijay 0 Feb 27 22:57 script.sh
digvijay@digvijay-virtualbox:~/workspace/source$
```

13. Change the owner of `backup/delta.txt` to the user `intern`.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/backup$ ls -l delta.txt
-rw-rw-r-- 1 digvijay digvijay 0 Feb 27 22:36 delta.txt
digvijay@digvijay-virtualbox:~/workspace/backup$ sudo chown intern delta.txt
digvijay@digvijay-virtualbox:~/workspace/backup$ ls -l delta.txt
-rw-rw-r-- 1 intern digvijay 0 Feb 27 22:36 delta.txt
digvijay@digvijay-virtualbox:~/workspace/backup$
```

14. Change the group ownership of the `workspace` directory to `training`.

```
digvijay@digvijay-virtualbox: ~
digvijay@digvijay-virtualbox:~$ ls -l workspace/
total 8
drwxrwxr-x 2 digvijay digvijay 4096 Feb 27 22:42 backup
drwxrwxr-x 2 digvijay digvijay 4096 Feb 27 22:57 source
digvijay@digvijay-virtualbox:~$ sudo chgrp -R training workspace/
digvijay@digvijay-virtualbox:~$ ls -l workspace/
total 8
drwxrwxr-x 2 digvijay training 4096 Feb 27 22:42 backup
drwxrwxr-x 2 digvijay training 4096 Feb 27 22:57 source
digvijay@digvijay-virtualbox:~$
```

15. Delete the file `source/beta.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace/source
digvijay@digvijay-virtualbox:~/workspace/source$ ls
beta.txt  gamma.txt  private.txt  public.txt  script.sh
digvijay@digvijay-virtualbox:~/workspace/source$ rm beta.txt
digvijay@digvijay-virtualbox:~/workspace/source$ ls
gamma.txt  private.txt  public.txt  script.sh
digvijay@digvijay-virtualbox:~/workspace/source$
```

16. Delete the `source` directory and all files remaining inside it using a single command.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ ls
backup  source
digvijay@digvijay-virtualbox:~/workspace$ rm -r source/
digvijay@digvijay-virtualbox:~/workspace$ ls
backup
digvijay@digvijay-virtualbox:~/workspace$
```

17. List all files in the `/etc` directory and save the output to a file named `etc_list.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:/$ ls -l etc > ~/workspace/etc_list.txt
digvijay@digvijay-virtualbox:/$ cd ~/workspace/
digvijay@digvijay-virtualbox:~/workspace$ ls
backup  etc_list.txt
digvijay@digvijay-virtualbox:~/workspace$ cat etc_list.txt
total 1128
-rw-r--r--  1 root root     3444 Jul  5  2023 adduser.conf
-rw-r--r--  1 root root       44 Feb 26 23:22 adjtime
drwxr-xr-x  3 root root     4096 Aug  5  2025 alsa
drwxr-xr-x  2 root root     4096 Feb 26 23:24 alternatives
-rw-r--r--  1 root root      335 Apr  8  2024 anacrontab
drwxr-xr-x  5 root root     4096 Aug  5  2025 apm
drwxr-xr-x  2 root root     4096 Aug  5  2025 apparmor
drwxr-xr-x  9 root root     4096 Aug  5  2025 apparmor.d
drwxr-xr-x  3 root root     4096 Aug  5  2025 apport
drwxr-xr-x  8 root root     4096 Feb 26 23:23 apt
drwxr-xr-x  3 root root     4096 Aug  5  2025 avahi
-rw-r--r--  1 root root     2319 Mar 31  2024 bash.bashrc
-rw-r--r--  1 root root       45 Jan 25  2020 bash_completion
```

18. Use a command to see the last 10 lines of the `/etc/passwd` file.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ tail /etc/passwd
saned:x:113:116::/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117::/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:113::/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
polkitd:x:989:989:User for polkitd:/:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
nm-openvpn:x:119:121:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
digvijay:x:1000:1000:Digvijay:/home/digvijay:/bin/bash
intern:x:1001:1001:intern,,,:/home/intern:/bin/bash
digvijay@digvijay-virtualbox:~/workspace$
```

19. Use a command to see the first 5 lines of the `/etc/group` file.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ head -n 5 /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,digvijay
digvijay@digvijay-virtualbox:~/workspace$
```

20. Search for the string "nologin" inside the `/etc/passwd` file and count how many lines contain it.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ grep "nologin" /etc/passwd
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112::/nonexistent:/usr/sbin/nologin
cups-pk-helper:x:111:113:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:990:990:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116::/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117::/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:113::/nonexistent:/usr/sbin/nologin
polkitd:x:989:989:User for polkitd:/:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
nm-openvpn:x:119:121:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
digvijay@digvijay-virtualbox:~/workspace$
```

21. Find all files in the `/var` directory that end with the extension `.log`.

```
digvijay@digvijay-virtualbox: /var
digvijay@digvijay-virtualbox:/$ ls
bin                boot  etc   lib    lib.usr-is-merged  media  opt   root  sbin               snap  swapfile  tmp  var
bin.usr-is-merged  dev   home  lib64  lost+found         mnt    proc  run   sbin.usr-is-merged  srv   sys       usr
digvijay@digvijay-virtualbox:/$ cd var/
digvijay@digvijay-virtualbox:/var$ sudo find /var -name "*.log"
[sudo] password for digvijay:
/var/log/boot.log
/var/log/gpu-manager.log
/var/log/installer/casper.log
/var/log/apt/history.log
/var/log/apt/term.log
/var/log/unattended-upgrades/unattended-upgrades-shutdown.log
/var/log/wvdialconf.log
/var/log/dpkg.log
/var/log/alternatives.log
/var/log/bootstrap.log
/var/log/apport.log
/var/log/Xorg.0.log
/var/log/auth.log
/var/log/kern.log
/var/log/sddm.log
/var/log/fontconfig.log
digvijay@digvijay-virtualbox:/var$
```

22. Create a symbolic link in your home directory named `link_to_delta` that points to `workspace/backup/delta.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace/backup
digvijay@digvijay-virtualbox:~/workspace/backup$ ln -s ~/workspace/backup/delta.txt ~/list_to_delta
digvijay@digvijay-virtualbox:~/workspace/backup$ ls -l ~/list_to_delta
lrwxrwxrwx 1 digvijay digvijay 41 Feb 27 23:39 /home/digvijay/list_to_delta -> /home/digvijay/workspace/backup/delta.txt
digvijay@digvijay-virtualbox:~/workspace/backup$
```

23. Create a hidden file named `.secret` in your home directory.

```
digvijay@digvijay-virtualbox: ~
digvijay@digvijay-virtualbox:~$ touch .secret
digvijay@digvijay-virtualbox:~$ ls -a
.                bda         list_to_delta  .secret                                  .vboxclient-display-svga-x11-tty2-control.pid  .vboxclient-seamless-tty2-service.pid
..               .cache      .local         snap                                     .vboxclient-display-svga-x11-tty2-service.pid  .vboxclient-vmsvga-session-tty2-control.pid
Assignment       .config     Music          .sudo_as_admin_successful                .vboxclient-draganddrop-tty2-control.pid       Videos
.bash_history    Desktop     Pictures       Templates                                .vboxclient-draganddrop-tty2-service.pid       workspace
.bash_logout     Documents   .profile       .vboxclient-clipboard-tty2-control.pid   .vboxclient-hostversion-tty2-control.pid       .xsession-errors
.bashrc          Downloads   Public         .vboxclient-clipboard-tty2-service.pid   .vboxclient-seamless-tty2-control.pid
digvijay@digvijay-virtualbox:~$
```

24. List all files in your home directory, including hidden files, in a long format ($ls -la$).

```
digvijay@digvijay-virtualbox: ~
digvijay@digvijay-virtualbox:~$ ls -la
total 1012
drwxr-x--- 17 digvijay digvijay   4096 Feb 27 23:40 .
drwxr-xr-x  4 root     root       4096 Feb 27 22:43 ..
drwxrwxr-x  3 digvijay digvijay   4096 Feb 27 21:03 Assignment
-rw-------  1 digvijay digvijay   1224 Feb 27 22:22 .bash_history
-rw-r--r--  1 digvijay digvijay    220 Mar 31  2024 .bash_logout
-rw-r--r--  1 digvijay digvijay   3771 Mar 31  2024 .bashrc
drwxrwxr-x  2 digvijay digvijay   4096 Feb 26 23:35 bda
drwxrwxr-x  9 digvijay digvijay   4096 Feb 27 21:13 .cache
drwxrwxr-x 10 digvijay digvijay   4096 Feb 26 23:35 .config
drwxrwxr-x  2 digvijay digvijay   4096 Feb 27 20:58 Desktop
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Documents
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Downloads
lrwxrwxrwx  1 digvijay digvijay     41 Feb 27 23:39 list_to_delta -> /home/digvijay/workspace/backup/delta.txt
drwxrwxr-x  4 digvijay digvijay   4096 Feb 26 23:26 .local
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Music
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Pictures
-rw-r--r--  1 digvijay digvijay    807 Mar 31  2024 .profile
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Public
-rw-rw-r--  1 digvijay digvijay      0 Feb 27 23:40 .secret
drwx------  4 digvijay digvijay   4096 Feb 27 22:16 snap
-rw-r--r--  1 digvijay digvijay      0 Feb 26 23:27 .sudo_as_admin_successful
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Templates
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-clipboard-tty2-control.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-clipboard-tty2-service.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-display-svga-x11-tty2-control.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-display-svga-x11-tty2-service.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-draganddrop-tty2-control.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-draganddrop-tty2-service.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-hostversion-tty2-control.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-seamless-tty2-control.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-seamless-tty2-service.pid
-rw-r-----  1 digvijay digvijay      5 Feb 27 20:57 .vboxclient-vmsvga-session-tty2-control.pid
drwxr-xr-x  2 digvijay digvijay   4096 Feb 26 23:26 Videos
drwxrwxr-x  3 digvijay training   4096 Feb 27 23:16 workspace
-rw-------  1 digvijay digvijay 904400 Feb 27 23:41 .xsession-errors
digvijay@digvijay-virtualbox:~$
```

25. Display the current system date and time and append it to a file named `timestamp.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ ls
backup  etc_list.txt
digvijay@digvijay-virtualbox:~/workspace$ date > timestamp.txt
digvijay@digvijay-virtualbox:~/workspace$ ls
backup  etc_list.txt  timestamp.txt
digvijay@digvijay-virtualbox:~/workspace$ cat timestamp.txt
Friday 27 February 2026 11:44:47 PM IST
digvijay@digvijay-virtualbox:~/workspace$
```

26. Display the name of the current logged-in user.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ who
digvijay tty2         2026-02-27 20:57 (:0)
digvijay@digvijay-virtualbox:~/workspace$ whoami
digvijay
digvijay@digvijay-virtualbox:~/workspace$ logname
digvijay
digvijay@digvijay-virtualbox:~/workspace$ echo $USER
digvijay
digvijay@digvijay-virtualbox:~/workspace$
```

27. Display the path of your current working directory.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ pwd
/home/digvijay/workspace
digvijay@digvijay-virtualbox:~/workspace$
```

28. Use the `df` command to check disk space in human-readable format and save it to `disk_usage.txt`.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ df
Filesystem     1K-blocks    Used Available Use% Mounted on
tmpfs             813324    1312    812012   1% /run
/dev/sda1       25621472 7786756  16507884  33% /
tmpfs            4066604       0   4066604   0% /dev/shm
tmpfs               5120       8      5112   1% /run/lock
tmpfs             813320     120    813200   1% /run/user/1000
digvijay@digvijay-virtualbox:~/workspace$ df > disk_usage.txt
digvijay@digvijay-virtualbox:~/workspace$ cat disk_usage.txt
Filesystem     1K-blocks    Used Available Use% Mounted on
tmpfs             813324    1312    812012   1% /run
/dev/sda1       25621472 7786760  16507880  33% /
tmpfs            4066604       0   4066604   0% /dev/shm
tmpfs               5120       8      5112   1% /run/lock
tmpfs             813320     120    813200   1% /run/user/1000
digvijay@digvijay-virtualbox:~/workspace$
```

29. Use the `du` command to check the size of the `workspace` directory.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ du ~/workspace/
4       /home/digvijay/workspace/backup
32      /home/digvijay/workspace/
digvijay@digvijay-virtualbox:~/workspace$
```

30. Identify your system's IP address using the command line.

```
digvijay@digvijay-virtualbox: ~/workspace
digvijay@digvijay-virtualbox:~/workspace$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fd17:625c:f037:2:e345:8901:bfe4:7dae  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::a2b0:c59f:9c53:8b9c  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:c843:b384:d250:c8fb  prefixlen 64  scopeid 0x0<global>
        ether 08:00:27:1b:ce:ff  txqueuelen 1000  (Ethernet)
        RX packets 124932  bytes 178741734 (178.7 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 10961  bytes 935081 (935.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 578  bytes 66409 (66.4 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 578  bytes 66409 (66.4 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

digvijay@digvijay-virtualbox:~/workspace$
```